

Genome-wide interaction reanalysis reveals context-specific lychee responses to *Peronophythora litchii*

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ABSTRACT

Background

Public lychee RNA-seq cohorts contain resistant/susceptible, tissue, time, and infection contrasts, but prior selected-gene analyses did not provide a genome-wide cultivar-by-infection discovery coupled to a prospective external-evidence firewall.

Methods

We locked PRJNA830488 as discovery and fitted a genome-wide negative-binomial model with cultivar, infection, and cultivar-by-infection terms. Candidates required genome-wide Benjamini-Hochberg $q < 0.05$, absolute interaction $\log_2$ fold change at least $\log_2(1.5)$, and uniform mapping/model quality. We froze genes, signed weights, pathways, and conditional transcript-usage results before opening external outcomes. Internal robustness covered independent quantification, edgeR, expression-filter sensitivity, leave-one-library-out analyses, and observed mapping sensitivity. Frozen evaluation used PRJNA450886 as the primary cross-context study; PRJNA922966 and PRJNA1090613 retained transfer or exploratory roles. Annotation, promoter motifs, small-RNA eligibility, and accession-aware literature were maintained as separate orthogonal classes.

Results

All 12 discovery libraries passed the technical inclusion gates. Among 19445 expressed genes, 262 met the genome-wide interaction threshold and 206 remained after uniform mapping/model QC. 16 frozen genes and 1 frozen pathway passed all internal robustness gates. In the locked 24-h PRJNA450886 evaluation, 2 genes and 0 pathways were cross-context supported, while 5 genes were contradictory. The frozen signature result was estimate 0.109, 95% CI -0.079 to 0.297, $q = 0.349$. Deterministic integration produced 0 Tier A genes, 12 Tier B genes, and 0 Tier C pathways; 65 entities were retired.

Conclusions

The evidence layers identify the exact scope of computational discoveries that are internally stable and, where observed, transport across a different tissue and resistant comparator. Cross-context support is not direct replication, and neither sequence annotation nor motif evidence establishes causality.

31 Keywords: lychee; *Litchi chinensis*; *Peronophythora litchii*; RNA-seq; genome-wide interaction;
32 cross-context validation; transcript usage; reproducibility

33 INTRODUCTION

34 Lychee (*Litchi chinensis*) production is constrained by downy blight caused by the oomycete
35 *Peronophythora litchii*. Public transcriptomic studies have profiled infection responses across
36 cultivars, tissues, and time points, creating an opportunity to distinguish genome-wide cultivar-
37 dependent responses from broad infection effects.

38 The principal discovery series, GSE201243/PRJNA830488, contains Guiwei and Yurong1
39 leaf libraries under mock and infected conditions at 24 h. Earlier interpretation of this accession
40 emphasized selected genes and promoter elements. Such selected-feature analyses cannot
41 control a genome-wide multiplicity family and risk using the same cohort for both selection and
42 apparent confirmation.

43 We therefore implemented a discovery/validation firewall. Dataset roles, models, thresholds,
44 empty-set behavior, and evidence vocabulary were frozen prospectively. Discovery refers only
45 to the locked genome-wide interaction in PRJNA830488; robustness refers only to sensitivity
46 within that study; external evidence is cross-context unless the biological estimand is directly
47 replicated; and orthogonal evidence remains separate from expression support.

48 RELATED WORK

49 Sun et al. analyzed Guiwei and Heiye pericarp at 6, 24, and 48 h (PRJNA450886;
50 doi:10.1038/s41598-019-39100-w) and reported distinct early response programs. Because that
51 accession is the present primary external cohort, its paper is prior interpretation rather than
52 independent validation. The GSE222650/651 Feizixiao small-RNA and mRNA studies provide
53 condition-matched tissue contrasts but no shared BioSample accessions, so they cannot
54 support specimen-level pairing. Molecular studies of the PIAvh202 effector
55 (doi:10.1093/plphys/kiad311) and a pathogen transcriptome (doi:10.1371/journal.pone.0178245)
56 provide biological context without automatically supporting a specific frozen host gene.

57 MATERIALS AND METHODS

58 Study design and evidence definitions

59 The protocol and amendments are in analysis/preregistration. PRJNA830488 was the only
60 discovery cohort. PRJNA450886 was primary cross-context evaluation, PRJNA922966 tested
61 generic tissue/infection transfer, PRJNA922965 was orthogonal small RNA, and

62 PRJNA1090613 remained exploratory because time and resistance metadata were unresolved.
63 No external result could alter frozen membership or weights.

64 **Dataset provenance and eligibility**

65 NCBI and GEO metadata were reconciled into a biological-unit registry. Inferential q values
66 are conditional on deposited-library independence because source-tree, pooling, harvest, and
67 extraction independence were not reported for the discovery libraries. The primary model
68 required at least three included libraries in every cultivar-treatment cell.

69 **References and preprocessing**

70 Host and pathogen references were combined with prefixed contig names. Paired reads
71 were checked with FastQC, trimmed with fastp, aligned by STAR two-pass mapping, counted
72 with featureCounts, and quantified independently with Salmon. Technical exclusion required
73 fewer than 10 million surviving pairs or less than 40% unique alignment. GenMap-based exon
74 uniqueness and observed candidate-level mapping checks were fixed before discovery.

75 **Discovery interaction model**

76 DESeq2 fitted counts with formula $\sim$ cultivar + treatment + cultivar:treatment, using Guiwei
77 and mock as reference levels. The primary coefficient was cultivarYurong1.treatmentinfected:
78 positive values indicate a stronger infected-minus-mock response in Yurong1 than Guiwei.
79 Genes required count ≥ 10 in at least three libraries, genome-wide BH $q < 0.05$, and $|\log_2 FC|$
80 $\geq \log_2(1.5)$. apeglm-shrunken interaction effects defined signed signature weights.

81 **Pathways and transcript usage**

82 Plant Reactome memberships were transported from one-to-one reciprocal-best Oryza
83 protein matches. Pathway discovery used the frozen mapping and genome-wide interaction
84 statistic. Transcript usage used Salmon abundances and a conditional inferential gate; no post
85 hoc substitute was permitted if the gate failed.

86 **Internal robustness**

87 Frozen genes were required to retain direction under Salmon gene aggregation, edgeR
88 quasi-likelihood testing, a CPM filter, and leave-one-library-out fits. Predeclared correlation,
89 effect-difference, q-value, leave-one-out, uniform mapping, and observed mapping-sensitivity
90 gates were combined conjunctively, without an additive score. Frozen pathways required
91 camera, roast, leading-edge deletion, and expression/length-matched null gates.

92 **External frozen evaluation**

93 PRJNA450886 used $\sim$ cultivar*treatment*time. The primary estimand was resistant-minus-
94 susceptible cultivar-by-infection at 24 h; 6 and 48 h were secondary. Frozen candidate tests
95 used BH adjustment across candidate-by-contrast families, an absolute $\log_2 FC$ threshold of

log₂(1.5), confidence intervals excluding zero, and discovery-direction agreement. PRJNA922966 used ~ tissue + treatment + tissue:treatment. PRJNA1090613 could not promote a final tier.

Annotation and orthogonal evidence

Representative proteins were selected by longest CDS with lexicographic transcript tie-breaking. Reviewed Swiss-Prot release 2026_02 DIAMOND matches, targeted InterPro/Pfam matches, and one-to-one Oryza orthology were separate classes. High-confidence computational annotation required at least two classes, ≥70% supported sequence coverage, and no detected architecture conflict.

The official PmiREN bulk archive was audited before small-RNA analysis. Its frozen gate was NOT_TESTABLE_REFERENCE_ABSENT; no study-derived novel-miRNA reference was substituted. Motif discovery used JASPAR 2026 Plants, strand-aware 1-kb and 2-kb promoters, 100 expression/GC-matched backgrounds, AME as primary, FIMO as sensitivity, and cognate-TF expression. Only discovery-frozen robust PWMs were transported to an independently selected PRJNA450886 response set.

Deterministic tiers and multiplicity

Tier A required gene discovery, complete internal robustness, primary cross-context support, no mapping/annotation failure, and at least one attributable orthogonal class. Tier B required discovery and robustness with partial or non-testable external evidence. Tier C was reserved for a robust, externally supported frozen pathway without a qualifying Tier A member. Contradictory direction or mapping/annotation failure retired an entity. Zero Tier A candidates was allowed.

Reproducibility

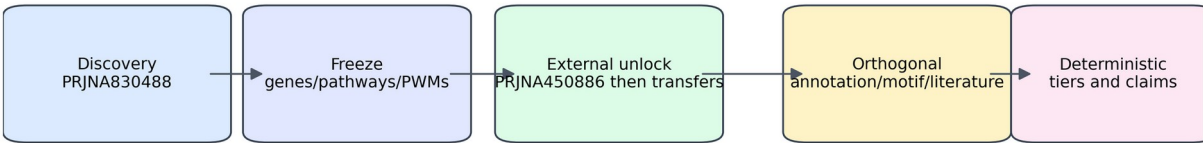
Snakemake workflows enforce staged discovery, external unlock, one STAR sorting job at a time, verified cleanup, and SHA-256 manifests. The resolved environment, synthetic fixtures, resource releases, all plotted source data, and protocol amendments accompany the results.

RESULTS

Provenance and discovery-data quality

All 12 PRJNA830488 libraries passed the fixed technical gate. The combined-reference alignment, per-library QC, PCA, sample distances, and uniform gene mappability are shown in Figure 2 and Supplementary S2.

Study design and discovery/validation firewall



Study	Locked role	Eligibility	Libraries
GSE201243	discovery	conditional_inferential	0
Sun_2019	primary_cross_context	cross_context	0
GSE222651	generic_infection_transfer	cross_context	0
GSE222650	orthogonal_modality	orthogonal	0
GSE262200	exploratory	exploratory	0

Figure 1. Locked dataset roles and discovery/validation firewall.

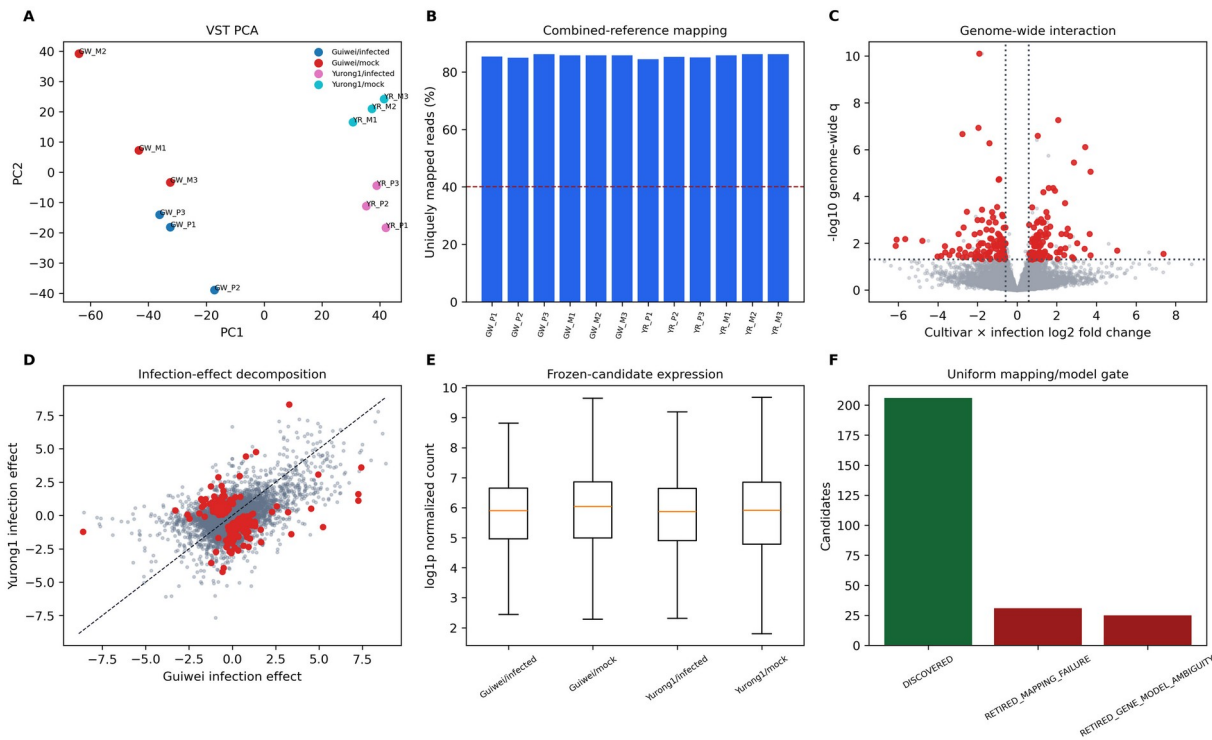


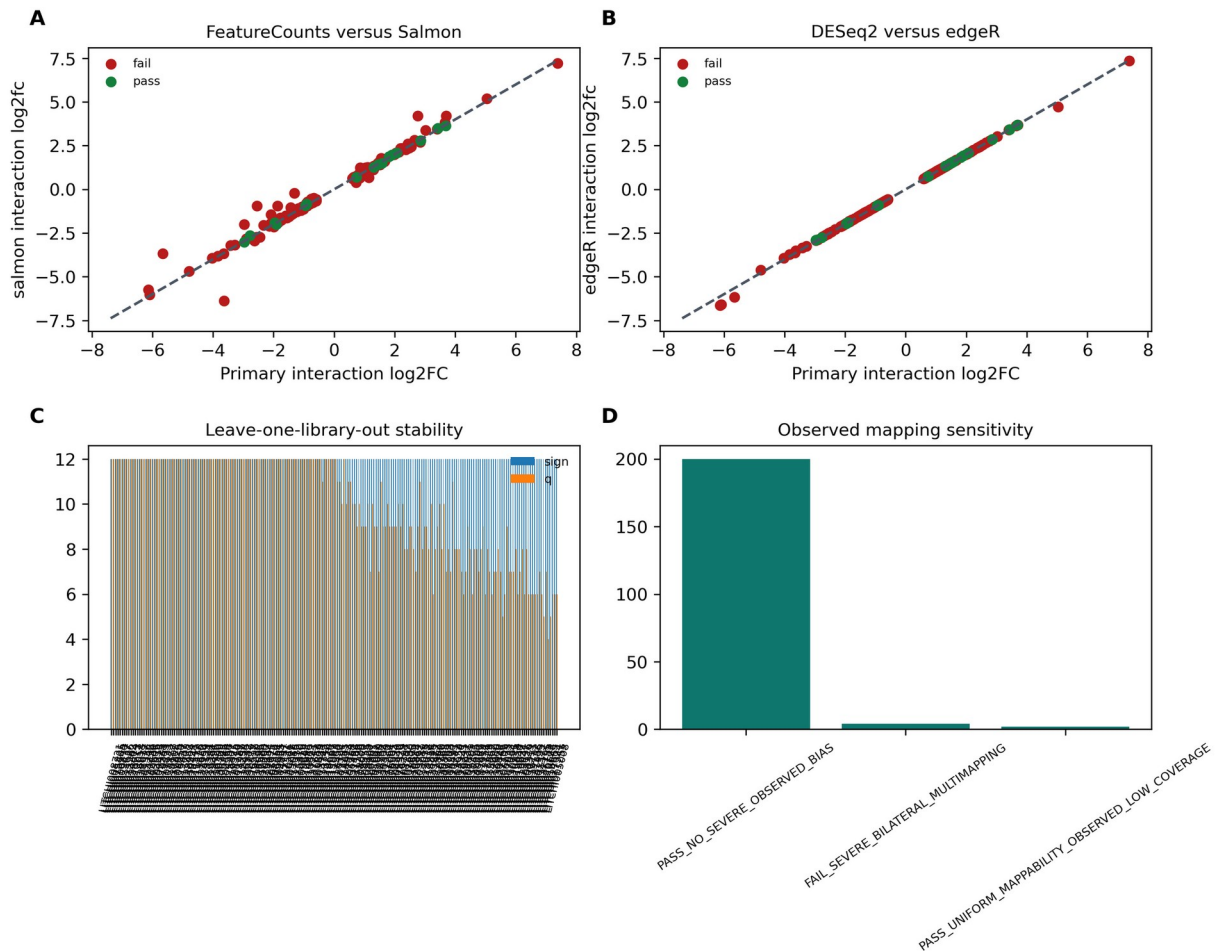
Figure 2. Discovery QC and genome-wide interaction.

Genome-wide cultivar-by-infection discovery

Of 19445 expressed genes, 262 passed the statistical interaction threshold before uniform mapping/model QC and 206 were frozen as discoveries. 6 pathways were frozen at the primary pathway threshold. Positive interaction effects indicate stronger infection response in Yurong1; negative effects indicate stronger response in Guiwei.

136 **Internal robustness**

137 16 of 206 frozen genes passed every quantification, statistical-method, filter, leave-one-out,
138 and mapping gate. 1 of 6 frozen pathways passed the four conjunctive pathway robustness
139 gates.



140
141 Figure 3. Independent internal robustness layers.

142 **Primary external cross-context evaluation**

143 At the locked PRJNA450886 24-h contrast, 2 frozen genes were cross-context supported, 5
144 were contradictory, and the remainder were unsupported or not testable under the fixed
145 threshold. The frozen signed-signature result was estimate 0.109, 95% CI -0.079 to 0.297,
146 $q=0.349$. No result is labeled direct replication because tissue and resistant comparator differ.

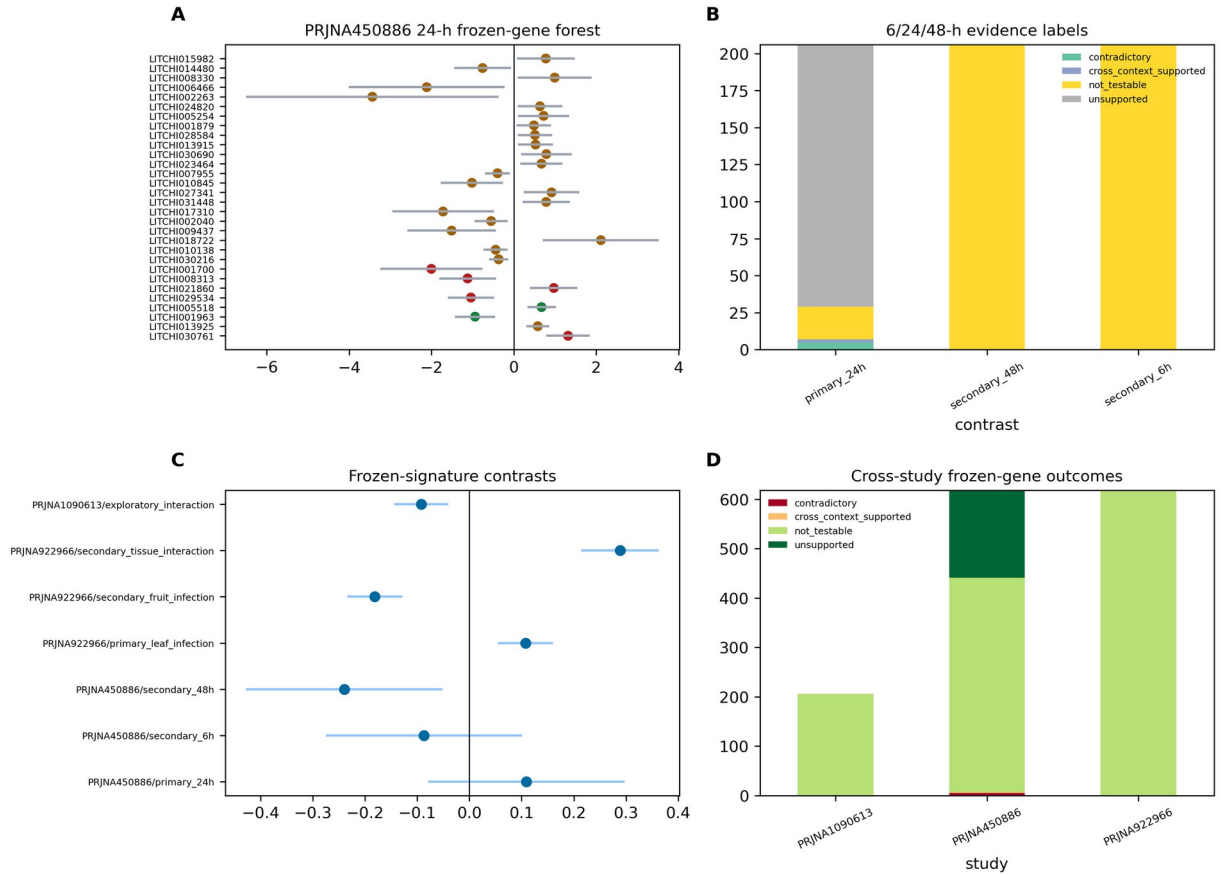


Figure 4. External frozen evaluation.

Frozen pathways and signature

0 frozen pathways passed camera, roast, fgsea, leading-edge deletion, and matched-null external gates in PRJNA450886. Transfer-study and exploratory outcomes are retained in electronic Table 4 and Supplementary S5-S6 without changing frozen definitions.

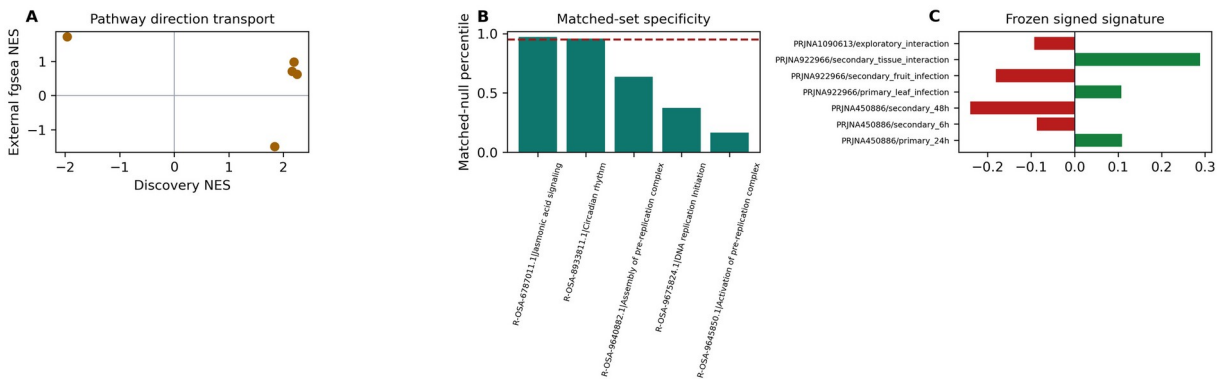


Figure 5. Frozen pathway and signed-signature evaluation.

Conditional transcript usage

225 frozen transcript-usage results passed the conditional gate.

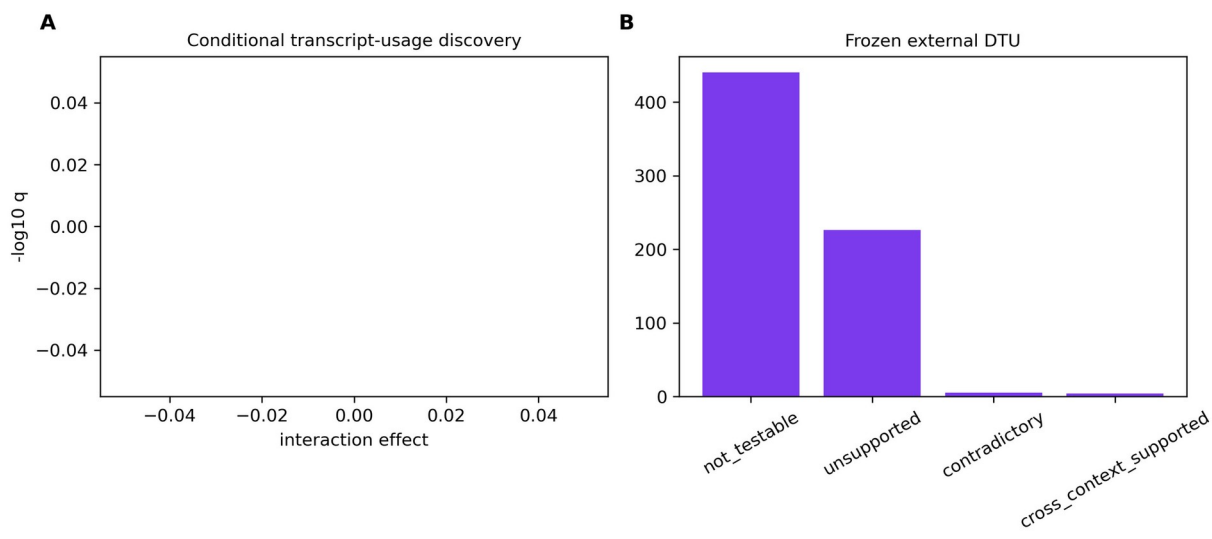


Figure 6. Conditional DTU discovery and external assessment.

Orthogonal evidence

0 frozen genes met the high-confidence computational annotation rule. 0 discovery motifs met the 100-background, two-window, and TF-expression gate; 0 of these transported under the external frozen test. Small-RNA coherence remained NOT_TESTABLE_REFERENCE_ABSENT. Literature entries reusing a current accession were retained as prior interpretation, not independent support.

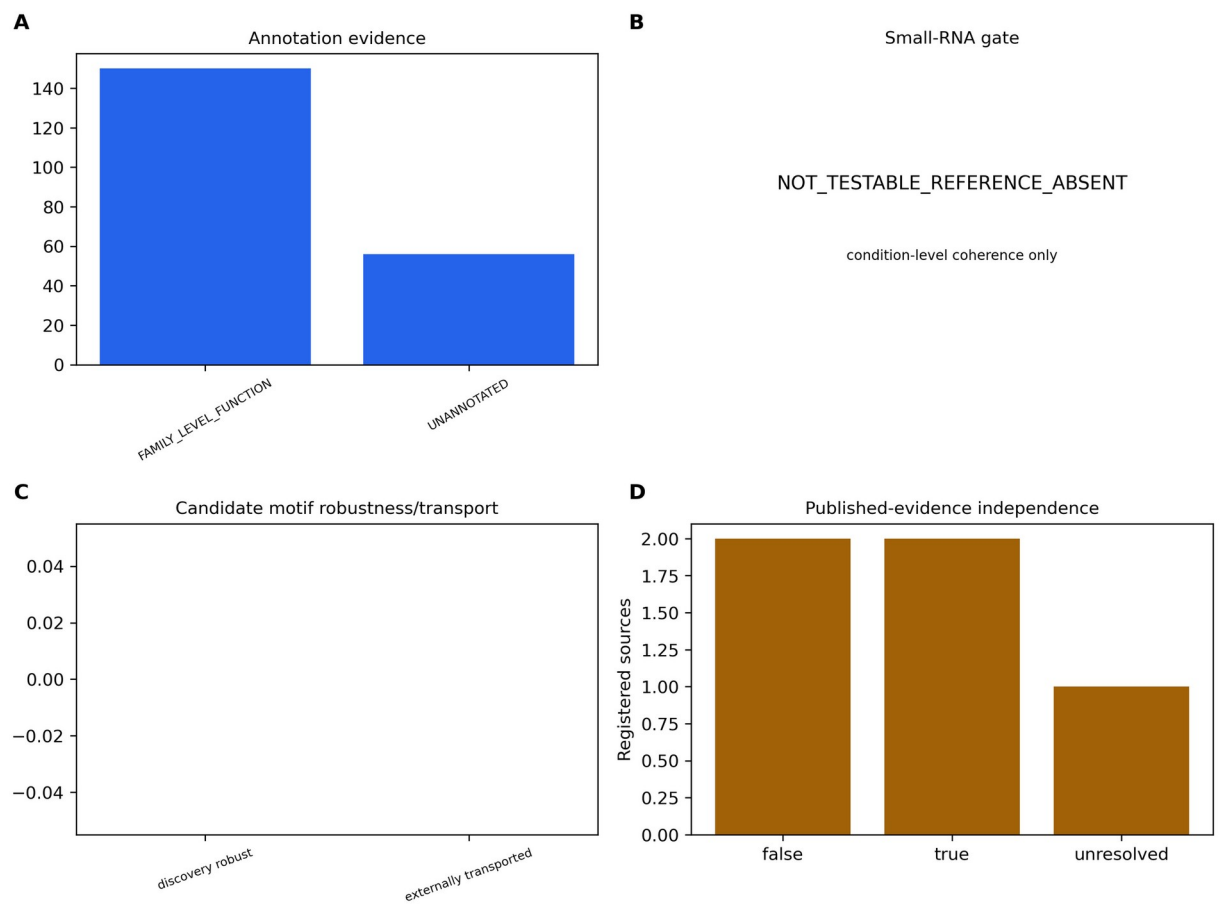
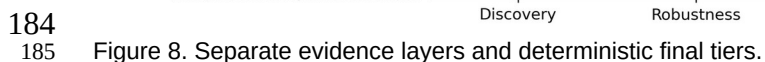


Figure 7. Orthogonal evidence layers.

Final deterministic tiers and contradictory results

The frozen matrix assigned Tier A to none, Tier B to LITCHI005805, LITCHI008721, LITCHI010877, LITCHI013915, LITCHI014613, LITCHI015740, LITCHI015982, LITCHI016077, LITCHI021546, LITCHI022605, LITCHI028717, LITCHI028926, and Tier C to none. 65 entities were retired, including LITCHI000241, LITCHI001085, LITCHI001279, LITCHI001696, LITCHI001700, LITCHI003064, LITCHI003379, LITCHI003665, LITCHI004137, LITCHI004429, LITCHI005026, LITCHI005338, LITCHI005404, LITCHI005631, LITCHI008313, LITCHI009437, LITCHI009978, LITCHI009986, LITCHI010009, LITCHI010490, LITCHI010704, LITCHI010779, LITCHI010874, LITCHI011561, LITCHI011966, LITCHI012300, LITCHI012927, LITCHI012980, LITCHI013400, LITCHI013576, LITCHI013662, LITCHI014006, LITCHI014435, LITCHI014512, LITCHI014699, LITCHI014858, LITCHI015541, LITCHI015623, LITCHI016030, LITCHI017118, LITCHI017811, LITCHI017972, LITCHI018449, LITCHI019399, LITCHI019568, LITCHI020348, LITCHI020381, LITCHI021593, LITCHI021860, LITCHI022640, LITCHI023325, LITCHI023626, LITCHI024633, LITCHI025853, LITCHI026893, LITCHI027558, LITCHI028829, LITCHI029455, LITCHI029534, LITCHI029642, LITCHI029701, LITCHI029848, LITCHI030155, LITCHI030447,

182 LITCHI030761. Contradictory and failed results remain visible in
183 results/candidates/contradictory_results.tsv.



186 **DISCUSSION**

187 The analysis replaces selected-gene reuse with a genome-wide interaction discovery whose
188 membership was fixed before external outcome access. This separation matters: a result can be
189 statistically discovered but internally unstable, internally robust but externally not testable, or
190 externally supported without being a direct replication.

191 PRJNA450886 changes tissue, resistant comparator, and time structure. Directionally
192 concordant threshold-passing results therefore demonstrate transport across context, not
193 repetition of the original biological experiment. PRJNA922966 addresses generic
194 tissue/infection transfer and cannot test the cultivar interaction. PRJNA1090613 remains
195 exploratory because unresolved metadata prevent a standardized confirmatory interpretation.

196 Pathway and signed-signature results can be more transportable than individual genes, but
197 they remain dependent on the frozen Oryza-to-lychee mapping and pathway universe. Leading-
198 edge deletion and matched-null analyses reduce, but do not eliminate, concerns about single-
199 gene dominance or generic expression/length effects.

200 Annotation, motif enrichment, and literature contribute different kinds of evidence.
201 Sequence/domain agreement supports a computational function label but not biological action in
202 infected lychee. A transported promoter PWM is a candidate regulatory pattern, not proof of
203 binding. The absent frozen PmiREN litchi reference made known-miRNA coherence non-
204 testable; substituting study-derived novel sequences after viewing the gate would have violated
205 the protocol.

206 Limitations include small deposited-library cell sizes, unknown source-unit independence,
207 possible reference bias despite uniform and observed mapping checks, and lack of a direct
208 biological replication cohort. External tissues and cultivar contrasts are deliberately
209 heterogeneous. The deterministic framework makes these limitations visible and permits zero
210 headline candidates rather than forcing a ranked list.

211 **CONCLUSION**

212 This study provides a genome-wide, prospectively frozen cultivar-by-infection reanalysis of
213 public lychee RNA-seq data, separates internal stability from cross-context evidence, and
214 releases the resulting candidate/signature resource with null and contradictory outcomes. The
215 final claims are computational and context-bounded; experimental perturbation and genuinely
216 independent biological replication remain necessary for causal inference.

217 **DATA AND CODE AVAILABILITY**

Primary public accessions are PRJNA830488/GSE201243, PRJNA450886, PRJNA922965/GSE222650, PRJNA922966/GSE222651, and PRJNA1090613/GSE262200. Exact reference releases and SHA-256 checksums are in analysis/config/resource_registry.tsv and data/reference manifests. The resolved environment is analysis/envs/lychee-discovery-resolved.yml. Protocol, amendments, result manifests, source data, and workflows are included in this repository. A repository release URL and archival DOI require the repository owner to publish the generated release and are explicitly marked pending in the submission-gate report.

SUPPLEMENTARY INFORMATION

Supplementary S1-S13 contain the biological-unit registry, per-library QC, all discovery statistics, robustness manifests, all external frozen tests, pathway/signature tests, DTU, annotation/orthology, small-RNA gates, motif/background tests, the evidence registry, scripts/environments/commands, and amendments. Every main figure has a TSV source-data file.

Electronic supplementary data

The complete machine-readable tables are distributed with the repository and are not embedded in this DOCX because several contain thousands to hundreds of thousands of rows. Every item is checksum-covered by the release bundle.

Electronic supplementary data inventory.

Artifact	Contents
results/tables/Table1_dataset_roles_eligibility.tsv	Dataset roles, estimands, eligibility, and locked limitations. Rows: 6.
results/tables/Table2_frozen_discovery.tsv	All frozen gene and pathway discoveries. Rows: 268.
results/tables/Table3_internal_robustness.tsv	Gene and pathway internal-robustness outcomes. Rows: 212.
results/tables/Table4_external_evaluation.tsv	All external frozen gene and pathway evaluations. Rows: 1,460.
results/tables/Table5_orthogonal_final_status.tsv	Orthogonal evidence and deterministic final status. Rows: 268.
results/supplement/S10_motif_background_tests.tsv	Motif/background tests and sensitivity results. Rows: 752,724.
results/supplement/S11_evidence_registry.tsv	Accession-aware published-evidence registry. Rows: 5.
results/supplement/S12_scripts_environments_commands.tsv	Scripts, environments, and command inventory. Rows: 62.
results/supplement/S13_amendment_deviation_log.tsv	Protocol amendments and deviations. Rows: 24.
results/supplement/S1_biological_unit_registry.tsv	Biological-unit and accession registry. Rows: 84.
results/supplement/S2_per_library_QC.tsv	Per-library discovery QC. Rows: 12.
results/supplement/S3_all_discovery_statistics.tsv	Genome-wide discovery statistics. Rows: 19,445.
results/supplement/S4_robustness_file_manifest.tsv	Internal robustness file manifest. Rows: 20.
results/supplement/S5_all_external_frozen_tests.tsv	All external frozen tests. Rows: 2,135.
results/supplement/S6_pathway_signature_tests.tsv	Pathway and signature tests. Rows: 25.
results/supplement/S7_DTU_results.tsv	Conditional differential-transcript-usage results. Rows: 15,790.
results/supplement/S8_annotation_orthology.tsv	Candidate annotation and orthology evidence. Rows: 206.
results/supplement/S9_small_RNA_results.tsv	Small-RNA reference-gate outcome. Rows: 1.

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237 **REFERENCES**

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239 resistant litchi cultivar. Scientific Reports. 2019. doi:10.1038/s41598-019-39100-w.
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245 5. JASPAR. <https://jaspar.elixir.no/>
246 6. InterPro. <https://www.ebi.ac.uk/interpro/>

247 **TABLES**

248 Table 1. Locked dataset roles and eligibility (condensed from electronic Table 1).

Accession	Design	Role	Eligibility
PRJNA830488	Guiwei/Yurong1 leaf; mock/infected; 24 h; 3 libraries/cell	discovery	conditional_inferential
PRJNA450886	Guiwei/Heiye fruit pericarp; mock/infected; 6/24/48 h; 3 libraries/cell	primary_cross_context	cross_context
PRJNA922966	Feizixiao leaf/fruit; mock/infected; 24 h; 3 libraries/cell	generic_infection_transfer	cross_context
PRJNA922965	Feizixiao leaf/fruit small RNA; mock/infected; 24 h; 3 libraries/cell	orthogonal_modality	orthogonal
PRJNA1090613	Guiwei/SFZ leaf; mock/infected; 3 libraries/cell	exploratory	exploratory

249

250 Table 2. Key validated computational outcomes.

Metric	Value
included discovery libraries	12
expressed genes tested	19445
statistical gene candidates	262
frozen gene discoveries	206
internally robust genes	16
frozen pathways	6
internally robust pathways	1
primary external cross context genes	2
primary external contradictory genes	5
primary external cross context pathways	0
frozen DTU rows	225
high confidence annotations	0
robust discovery motifs	0
externally transported motifs	0
Tier A	0
Tier B	12
Tier C	0

27

Metric	Value
Retired	65
contradictory or failed rows	65

251

252 Table 3. Deterministic final tier distribution.

Tier/status	Count
Tier A	0
Tier B	12
Tier C	0
Exploratory	191
Retired	65

253

254